Chain of Custody

1. Chain of Custody

Chain of custody refers to the chronological documentation or paper trail that records the sequence of custody, control, transfer, analysis, and disposition of physical or electronic evidence. It is a critical concept in legal contexts (such as criminal trials) and in fields like forensic science, cybersecurity, supply chain management, and environmental testing.

Key purposes:

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- **Preserve integrity –** Ensure the evidence has not been tampered with, altered, or substituted.
- **Establish authenticity –** Prove that the evidence presented in court is the same as what was originally collected.
- **Prevent contamination –** Especially crucial for DNA, chemical samples, or digital data.
- **Liability and compliance –** In supply chains (e.g., food or pharmaceuticals), it ensures product safety and regulatory compliance.

How it work

A typical chain of custody log includes:

- Unique identifier for the item (e.g., evidence number)
- Description of the item
- Date and time of each transfer
- Names and signatures of individuals who handled it
- Purpose of transfer (e.g., collection, storage, testing)
- Security measures (e.g., sealed containers, access logs)

Example in a criminal case:

1. Police officer collects a bloody knife at a crime scene, seals it in a bag, signs and dates it.
2. Key purposes:
3. Preserve integrity – Ensure the evidence has not been tampered with, altered, or substituted.
4. Establish authenticity – Prove that the evidence presented in court is the same as what was originally collected.
5. Prevent contamination – Especially crucial for DNA, chemical samples, or digital data.
6. Liability and compliance – In supply chains (e.g., food or pharmaceuticals), it ensures product safety and regulatory

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compliance.

If the chain is broken (e.g., missing signature, unsealed package), the evidence may be deemed inadmissible in court because its authenticity cannot be guaranteed.

2. **A legal record tracking evidence from collection to courtroom presentation**
   Chain of custody is that legal record. It is the verifiable, unbroken trail of documentation that tracks evidence from the moment it is collected at a crime scene or source, through every transfer, storage, analysis, and handling step, all the way to its presentation in court.

Why it's a legal record

- It's not just a note—it's an official document (often a log, form, or digital audit trail) that is subject to legal scrutiny.
- It must be complete, accurate, and contemporaneous (created at the time of each action, not later from memory).
- It can be challenged in court by opposing counsel. Any gap or inconsistency can be used to attack the evidence's credibility.

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What the record typically includes

Element	Example
Unique ID	"Exhibit A-1"
Description	"9mm Smith & Wesson pistol, serial #12345"
Collection info	Date, time, location, collector's name
Each handler	Name, title, signature, date/time received and released
Storage conditions	Locked evidence locker, temperature log (for biological samples)
Purpose of transfer	"For fingerprint analysis" or "To D.A.'s office for trial"
Final chain link	Introduction in court by a witness who identifies the item

The courtroom presentation link

- At trial, the prosecutor must call every person who handled the evidence (or at least show the documented chain) to testify that the item is in substantially the same condition as when collected.

- The judge will only admit the evidence if the chain of custody is unbroken and trustworthy.

- If defense counsel can show a break—e.g., an unlocked evidence room, a missing signature, a period where custody is unknown—the judge may exclude the evidence entirely.

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Real-world consequence

Without a proper chain of custody, even a bloody glove with the defendant's DNA and the victim's blood is useless in court—because the defense can argue it might have been planted, swapped, or contaminated after collection.

So your definition is spot on: a legal record tracking evidence from collection to courtroom **presentation**—and that record is what gives the evidence its legal power.

3. **A break in the chain can render evidence inadmissible in court**
   Why a break is so devastating
   The court's underlying assumption is: No chain, no proof of authenticity.
   If the prosecution cannot account for who had the evidence and what was done to it at every single moment, then the evidence could have been:

   - Planted (e.g., someone added the defendant's fingerprints later)
   - Switched (e.g., a clean weapon substituted for the real one)
   - Contaminated (e.g., DNA from another person introduced)
   - Tampered with (e.g., drugs diluted or replaced)
   - Degraded (e.g., blood sample left in a hot car)

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What counts as a break

Example of a break	Why it matters
Missing signature on transfer log	Unknown who held the evidence for 6 hours
Unsealed or torn evidence bag	Could have been opened and contents altered
Evidence locker left unlocked	Potential for unauthorized access
No log of who removed evidence for testing	Analyst might have swapped samples
Gaps in surveillance footage of storage area	Cannot confirm no tampering occurred

The legal standard

Courts don't require absolute perfection (minor administrative errors like a missing middle initial may be excused if the chain is still trustworthy). But a material break—where a reasonable person would doubt the evidence's integrity—leads to inadmissibility.

Example from case law

In United States v. Llera Plaza (2002), drug evidence was deemed inadmissible because the chain of custody log showed the evidence bag had been opened by an officer who was not documented as authorized to handle it. The break was small but fatal.

Practical result for trial

- If the judge rules evidence inadmissible due to a broken chain:
- The jury never sees or hears about that evidence.
- The prosecution may lose its strongest piece of proof.
- The case could be dismissed if no other evidence remains.

Draft a Mock Chain-of-Custody Form

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Chain of Custody

Below is a realistic template you can copy, print, or recreate. It includes all critical fields that real agencies (police, labs, corporate security) use.

Mock Chain of Custody Log

Case/Incident	**2026-04-002**	**Investigator:**	**Det. M. Rivers**
Item	1 of 3	Item Description:	Samsung Galaxy S23, black case, cracked screen, powered off
Collection Date/Time:	2026-04-02 09:15	Location Collected:	1427 Maple St., Bedroom nightstand
Collected By (Print/Sign):	Officer J. Chen / signed	Agency/Dept:	Patrol Division

Phases of Digital Forensic Investigation

Think of these as a linear workflow, though in practice you may loop back (e.g., analysis reveals new evidence sources to identify).

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DIGITAL FORENSIC ANALYSIS REPORT
1. Case Information

Case Name: sh90
Case Number: shu90
Examiner: Shubham
Date of Analysis: 03-Apr-2026
Tools Used: Autopsy, FTK Imager, CertUtil

2. Objective

The objective of this forensic investigation is to analyze a disk image file and identify relevant digital evidence including deleted files, sensitive data, and system artifacts while maintaining evidence integrity.

3. Evidence Description

The primary evidence analyzed in this case is a forensic disk image file:

- **Image File:** Datarecover.E01
- **Type:** EnCase Image Format (E01)
- **Source:** USB/Test Drive

The image was created using FTK Imager following standard forensic procedures.

4. Evidence Integrity Verification

To ensure that the evidence has not been altered, SHA-256 hashing was performed.

Hash Tool Used: CertUtil (Windows)

Command Used:

```
certutil -hashfile Datarecover.E01 SHA256
```

Generated SHA-256 Hash:
6d43fe7c72d1452fa679c7d6aeab1e738a100312070fca48963025367a4fffa0

The hash value confirms that the integrity of the forensic image is maintained.

5. Methodology
6. Analysis Using Autopsy

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6.1 Data Source Import

The disk image (Datarecover.E01) was successfully imported into Autopsy. The tool parsed the file system and identified partitions and volumes.

6.2 File System Analysis

- Multiple volumes detected
- Large portion of data found in unallocated space
- Files marked as "unknown" and "unallocated"

This indicates deletion or incomplete file allocation.

7. Deleted File Recovery

Autopsy recovered numerous deleted files from unallocated space.

Recovered File Types:

- Images: PNG, JPG
- Documents: PDF

Notable Files:

- debian.jpg
- digital meter.pdf
- aviation forensic.pdf
- splash.png

These files indicate previous user activity and possible relevance to the investigation.

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8. Slack Space Analysis

Slack space analysis revealed residual data in:

- WPSettings.dat-slack
- IndexerVolumeGuid

Slack space may contain remnants of previously deleted or modified data, making it valuable forensic evidence.

Relevant matches may indicate sensitive or confidential information stored within the system.

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9. Timeline Analysis

Timeline analysis helps reconstruct system activity.

Findings include:

- File creation timestamps
- File modification timestamps
- Evidence of deletion events

This helps determine when specific actions occurred on the system.

10. Findings Summary

- Evidence integrity verified using SHA-256 hash
- Multiple deleted files successfully recovered
- Sensitive file types identified (images and PDFs)
- Significant data found in unallocated space
- Slack space contains residual hidden data
- Timeline indicates user activity and file deletion events

11. Conclusion

The forensic analysis confirms that the disk image contains recoverable deleted files and residual data in slack space. The presence of documents and images suggests prior user activity. Evidence integrity has been maintained throughout the investigation. The findings may be useful for further investigation or legal proceedings.

12. Recommendations

- Perform deeper keyword analysis for sensitive information
- Use additional tools for artifact extraction (browser history, logs)
- Correlate timeline data with user activity
- Preserve all evidence securely for legal use

13. Appendix

- Hash Value: 6d43fe7c72d1452fa679c7d6aeab1e738a100312070fca48963025367a4fffa0
- Tools Used: Autopsy, FTK Imager, CertUtil
- Evidence Location: D:\Evidence\Case001

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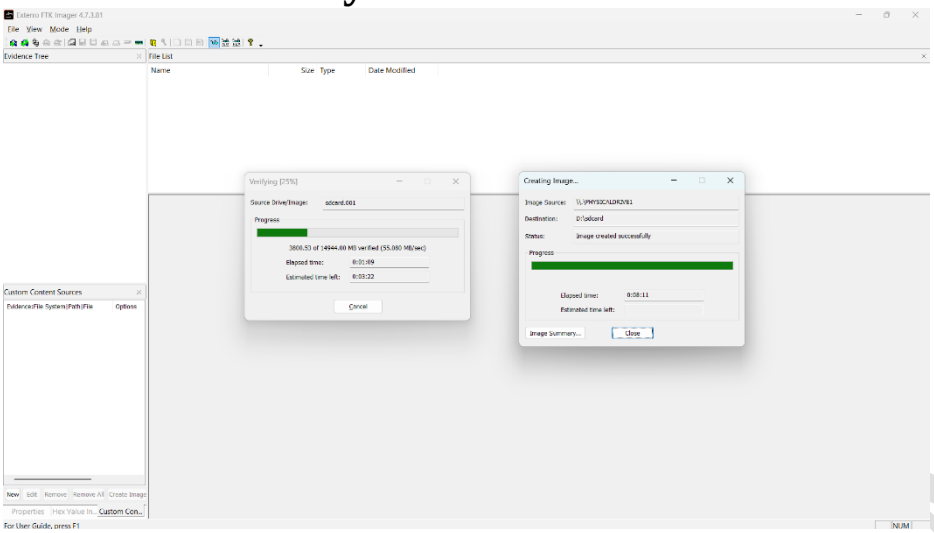

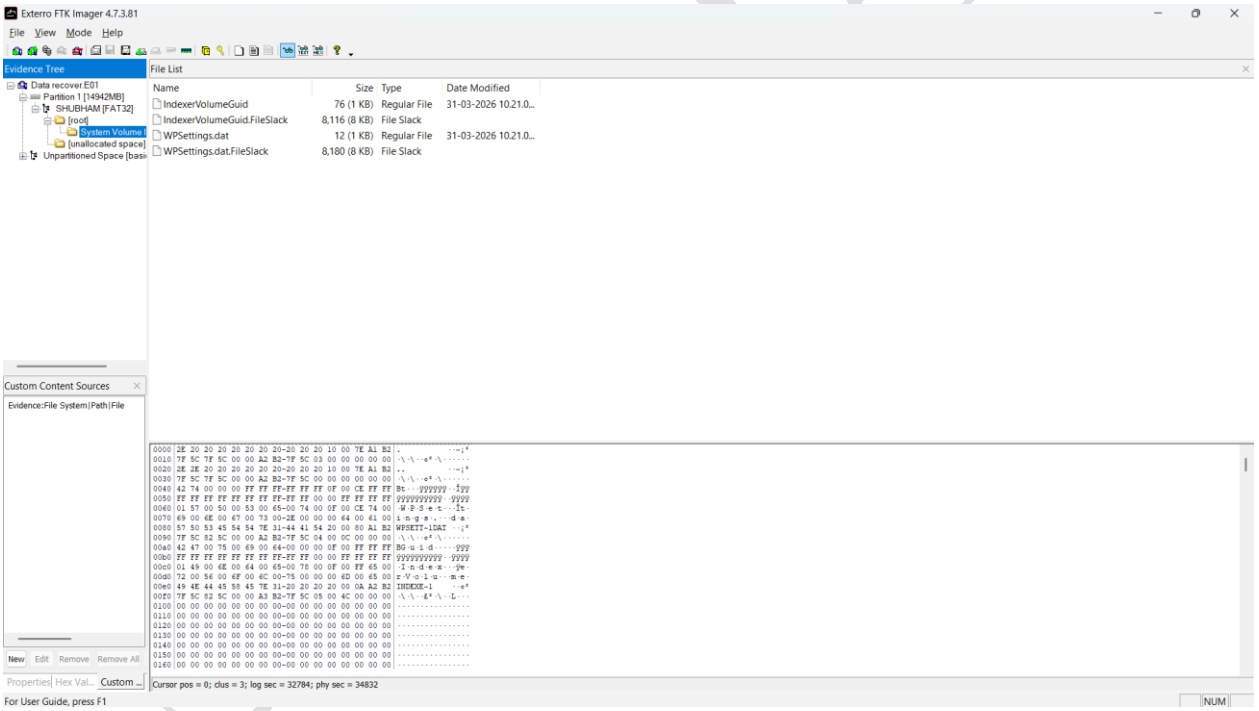

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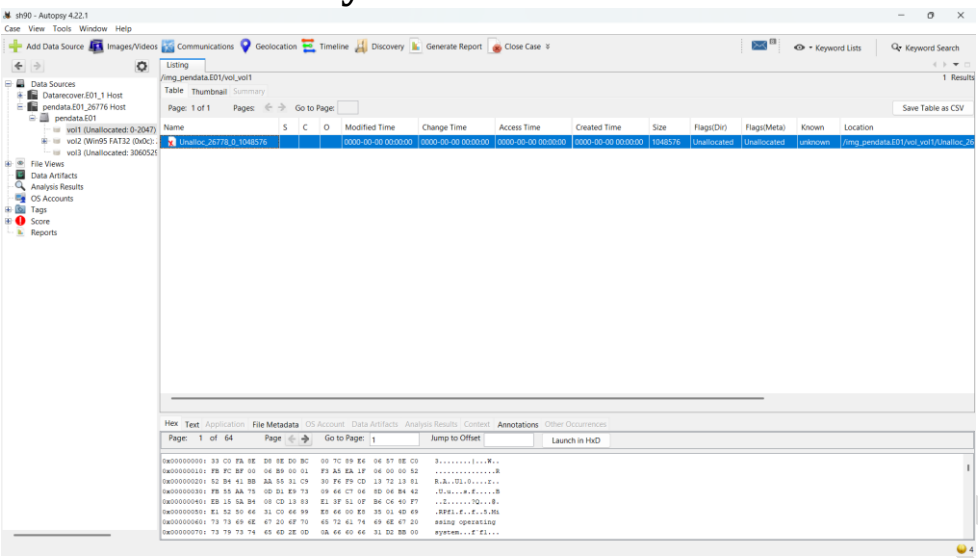

```
PS D:\Ftkimager> CertUtil -hashfile Datarecover.E01 SHA256
SHA256 hash of Datarecover.E01:
6d43fe7c72d1452fa679c7d6aeab1e738a100312070fca48963025367a4fffa0
CertUtil: -hashfile command completed successfully.
PS D:\Ftkimager>
```

SIMULATION

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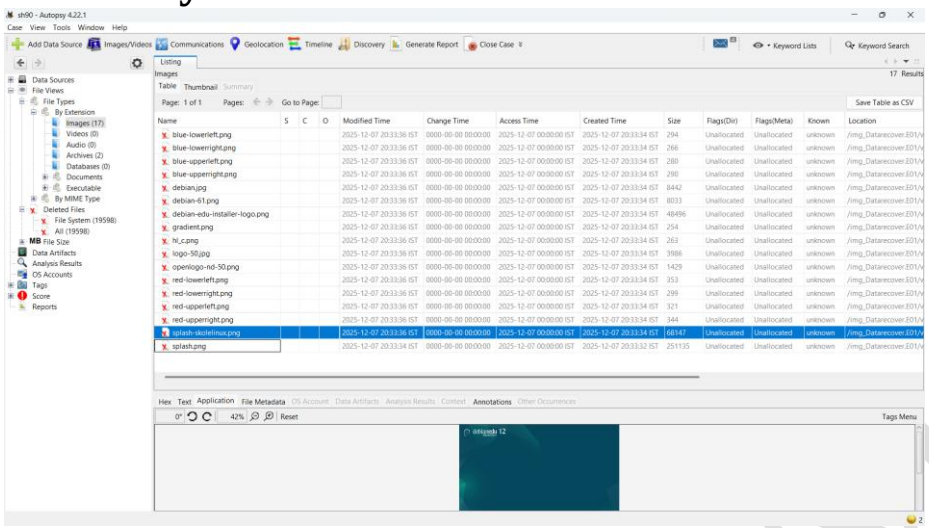

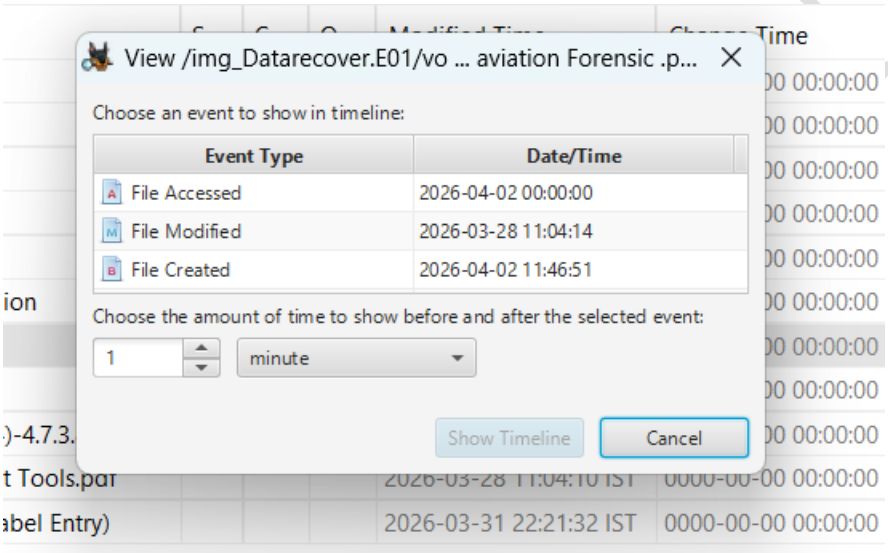

Summary		
Case Name:	sh90	
Case Number:	shu90	
Number of data sources in case:	2	
Examiner:	Shubham	

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Network Traffic Analysis with Wireshark

Wireshark is a software tool used to monitor the network traffic through a network interface. It is the most widely used network monitoring tool today. Wireshark is loved equally by system administrators, network engineers, network enthusiasts, network security professionals and black hat hackers.

The extent of its popularity is such, that experience with Wireshark is considered as a valuable/essential trait in a computer networking-related professional.

Introduction to Web Server

- A web server stores and delivers web content to users over the internet, primarily using HTTP or HTTPS, while an HTTP server specifically handles HTTP requests and responses.
    - Responds to browser requests by serving web resources.
    - HTTP server is a type of web server focused on HTTP communication.
    - Some web servers support additional protocols beyond HTTP.

Domain Name System (DNS)

DNS is a system that translates human-readable domain names into IP addresses so computers can communicate over the internet. It works in a hierarchical and distributed manner to efficiently resolve queries. DNS ensures users can access websites without remembering numeric IP addresses. It also improves performance using caching.

- Provides name-to-address resolution
- Uses a hierarchy (Root, TLD, Authoritative servers)
- Improves speed using caching mechanisms

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Internet Control Message Protocol (ICMP)

Internet Control Message Protocol (ICMP) is a network layer protocol widely used by network devices such as routers, gateways and hosts to send error messages and operational information. Since the Internet Protocol (IP) itself does not have an inbuilt error-reporting or correction mechanism, ICMP is a supporting protocol within the IP suite that helps in reporting errors and sending diagnostic messages. It is primarily used for:

- **Error reporting**: When data packets cannot reach their destination due to issues such as unreachable hosts, timeouts or fragmentation errors.
- **Operational queries**: For example, echo requests and replies used in tools like ping.

Uses of ICMP

Error Reporting

- If a message cannot be delivered, ICMP informs the source about the failure.
- Example: If a packet is too large and cannot be forwarded, the receiver drops the packet and sends an ICMP error message to the sender.

Network Diagnostics

- **Traceroute**: Used to determine the path packets take across routers to reach the destination.
- **Ping**: Sends echo-request and echo-reply messages to measure round-trip time and test connectivity.

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Suspicious Activity Analysis

The captured traffic was analyzed for potential threats:

Outbound Connections

- All outbound connections were directed to known and trusted servers
- No unexpected or malicious IP addresses identified

Port Analysis

- Traffic observed on standard ports:
  - 80 (HTTP)
  - 443 (HTTPS)
  - 53 (DNS)
- No unusual or suspicious ports detected

Traffic Patterns

- No abnormal traffic spikes or repeated connections observed
- No signs of malware communication or data exfiltration

Findings

- Network traffic was successfully captured and analyzed
- HTTP, DNS, and ICMP packets were identified using filters
- No suspicious activity or malicious connections were detected
- Network behavior appeared normal during the capture period

Conclusion

The Wireshark analysis demonstrated normal network activity within the monitored environment. All observed traffic corresponded to expected user actions such as web browsing and DNS resolution. No evidence of malicious communication, unauthorized access, or abnormal behavior was found. This confirms that the network environment was secure during the capture session.

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Recommendations

- Perform longer duration captures for deeper analysis
- Monitor network regularly for anomalies
- Use intrusion detection systems (IDS) for real-time alerts
- Analyze encrypted traffic (HTTPS) using advanced techniques if required

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Appendix

- Capture Duration: 5 minutes
- Protocols Analyzed: HTTP, DNS, ICMP
- Tool Used: Wireshark

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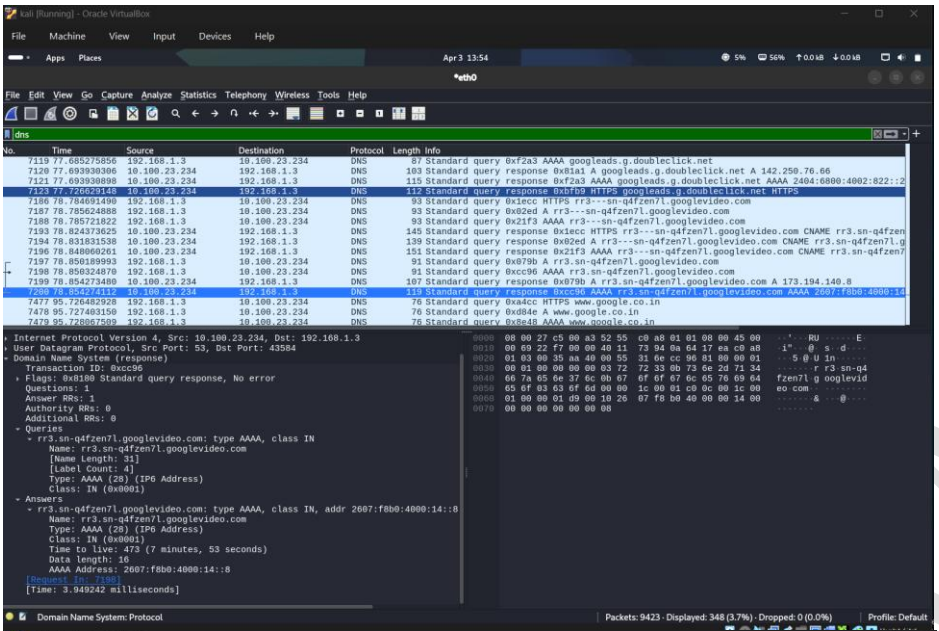

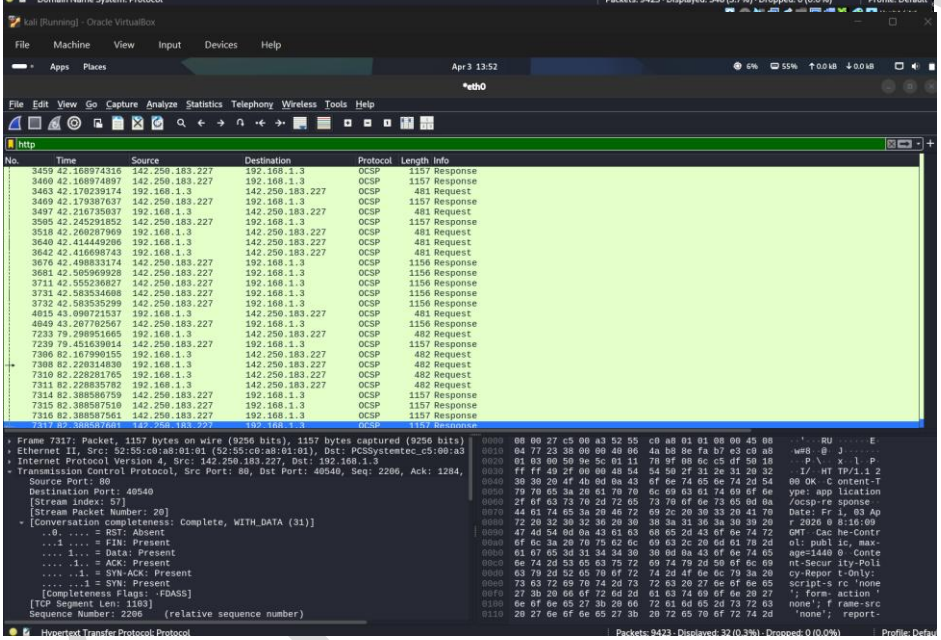

During the Wireshark capture, network traffic primarily consisted of normal HTTP, DNS, and ICMP packets generated through routine browsing and system activity. DNS queries resolved common domains, while HTTP requests showed standard web communication. ICMP packets indicated normal connectivity checks. No significant anomalies such as unusual outbound connections, suspicious IP addresses, or non-standard ports were detected. Overall, the network behavior appeared normal, with no evidence of malicious activity, data exfiltration, or unauthorized access during the observation period.

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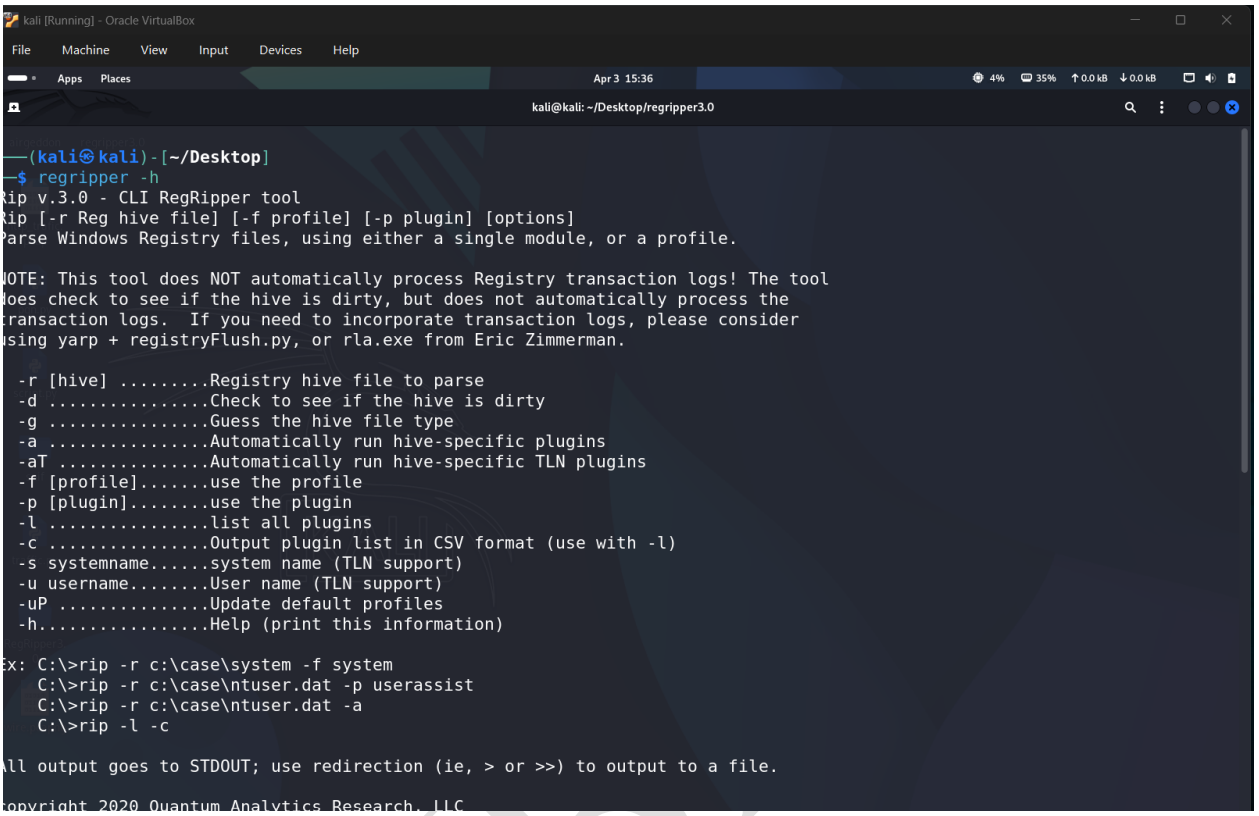
Registry Analysis with RegRipper



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```
kali [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
Apps  Places                Apr 3  15:37                2%  34%  0.0 kB  0.0 kB
kali@kali: ~/Desktop/regripper3.0

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┌──(kali㉿kali)-[~/Desktop]
└─$ sudo apt install git build-essential perl libparse-win32registry-perl -y
[sudo] password for kali:
git is already the newest version (1:2.53.0-1).
git set to manually installed.
build-essential is already the newest version (12.12).
build-essential set to manually installed.
perl is already the newest version (5.40.1-7).
libparse-win32registry-perl is already the newest version (1.1-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 112

┌──(kali㉿kali)-[~/Desktop]
└─$ git clone https://github.com/keydet89/regripper3.0.git
Cloning into 'regripper3.0'...
remote: Enumerating objects: 646, done.
remote: Counting objects: 100% (84/84), done.
remote: Compressing objects: 100% (58/58), done.
remote: Total 646 (delta 46), reused 28 (delta 26), pack-reused 562 (from 2)
Receiving objects: 100% (646/646), 5.64 MiB | 5.04 MiB/s, done.
Resolving deltas: 100% (328/328), done.

┌──(kali㉿kali)-[~/Desktop]
└─$ cd regripper3.0

┌──(kali㉿kali)-[~/Desktop/regripper3.0]
└─$ chmod +x rip.pl

┌──(kali㉿kali)-[~/Desktop/regripper3.0]
└─$ sudo ln -s $(pwd)/rip.pl /usr/local/bin/rr
```

- ✓ RegRipper could not run in my system and I could not understand 2 tasks (6,7) I am trying, I need more time, I will complete it, please give me more time

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